



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

FACULTY OF COMPUTING

SEMESTER 2/2022-2023

SECV2113 SECTION 01

HUMAN COMPUTER INTERACTION

Assignment 2

**FOODTM:
Easier Food Access on UTM campus**

HEURISTIC EVALUATION

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Introduction

In Assignment 2 of Human-Computer Interaction course, we will be conducting **heuristic evaluations** on the paper prototype that we have designed in Project Part 4 as part of the progression for the project in general. Here, we will apply **Nielsen's heuristics** to evaluate the designed user interfaces while rating them with the severity level based on observation.

Nielsen's Heuristics consists of 10 rules :

1. **Visibility of system status.** – Necessity of real time information providing
2. **Match between system and real world.** – Familiarity of System to user
3. **User control and freedom.** – Multifunction application
4. **Consistency and standards.** – Design matches expectation and is consistent
5. **Error prevention.** – Warning triggers for certain actions
6. **Recognition rather than recall.** – Recognition of Pages allows easier access
7. **Flexibility and efficiency of use.** – Shortcuts and efficient solutions to problems
8. **Aesthetic and minimalist design.** – Consideration of button design relevancy
9. **Help users recognize, diagnose, and recover from errors.** – Let user understand what went wrong
10. **Help and documentation.** – Guides and informations given to user

Severity Rating :

1. **Cosmetic Issue**
2. **Minor Issue**
3. **Major Issue**
4. **Catastrophic Issue**

Through the identification of issues of our paper prototype using these rules, we will be able to revise the designs and improve them to fit societal expectations hence inviting more users to our application upon launching.

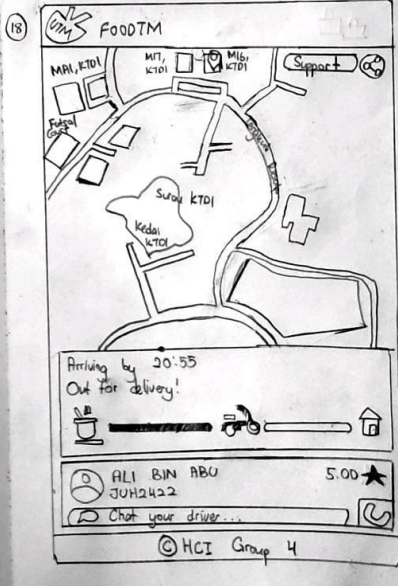
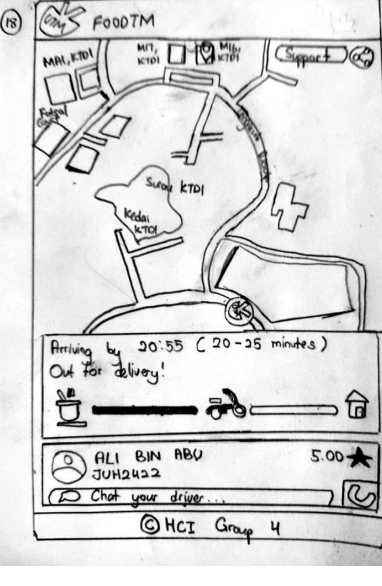
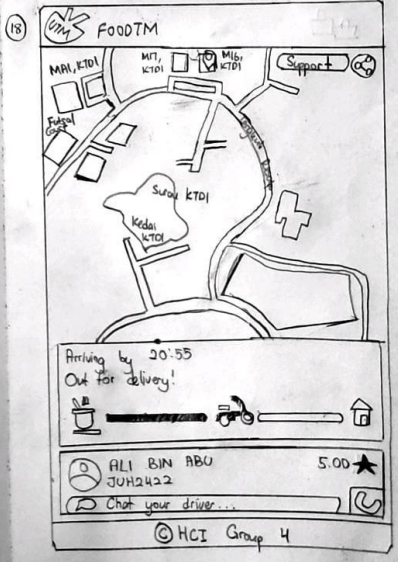
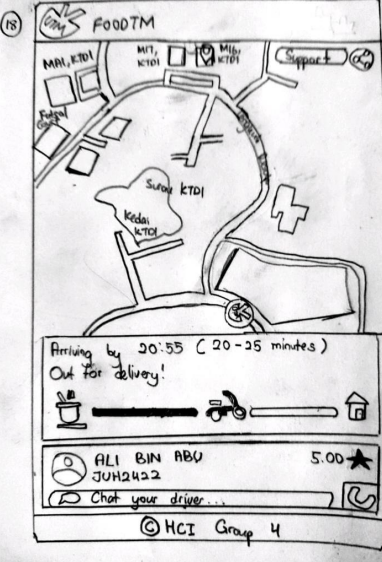
Method of evaluation

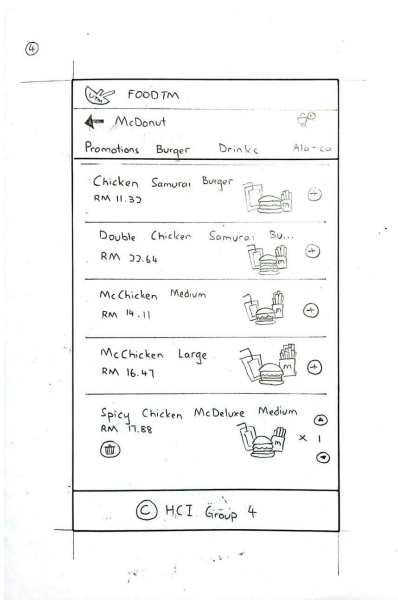
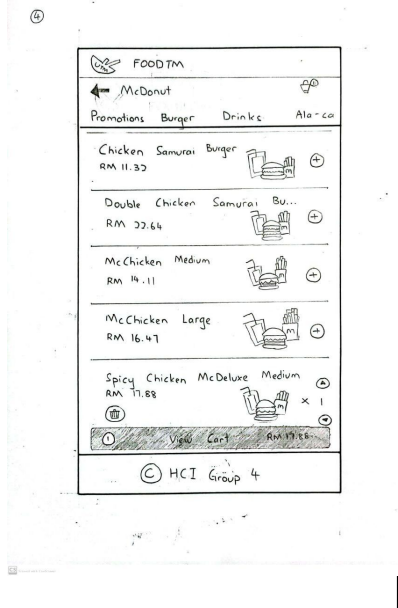
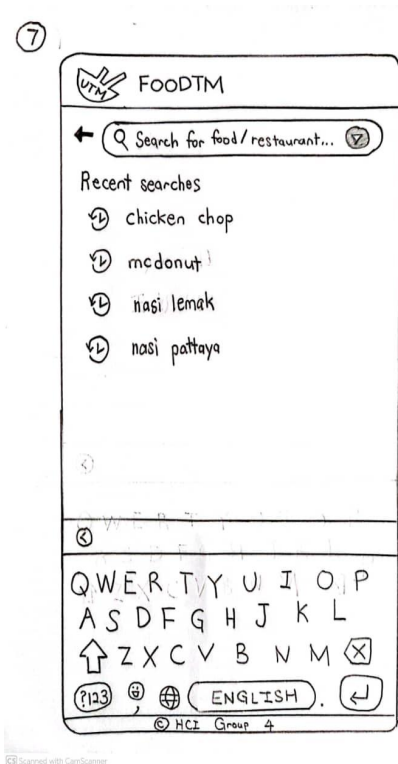
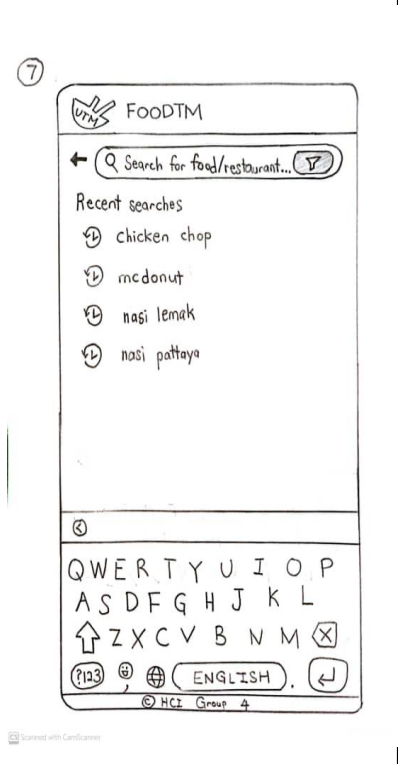
We will be carrying out the evaluation process by splitting roles among the group members. All 5 members will understand Heuristic Evaluation and then each test the paper prototypes one by one.

We will then gather all opinions from each group member to decide the violations existing in our application design and proceed to revise the design, based on the 10 rules by Nielsen's Heuristics. The result of the updated design will then be used to test on random users for further findings.

Result

No.	Prototype Image	Identified Issue	Heuristic & Severity	Revised Image	Revised Explanation
1		No confirmation before deleting the entire order (when the “trash can” button is pushed). Users need to search for the item again if required.	H5 S2		A confirmation interface for user to confirm the action to remove the item before the item is deleted
2		Keyword, word-to-word search, and delete are not detailed (only icons shown). Users might not be able to recognise the function based on the icons.	H10 S3		Text describing the functions “keyword”, “word-to-word search”, and “delete” are added besides their respective icons.

3		There is how much time it takes to arrive at the destination to deliver the food. Users may be unable to predict how long it will take to arrive at their destination.	H3,H5 S2		It displays the number of minutes until the food arrives at its destination. Users will be ready to eat their meals before the vehicle arrives at its destination.
4		The rider's location is not indicated on the map. Users will be perplexed and believe that the application contains bugs.	H3 S3		The rider's location is displayed on the map. Such visibility enables users to see where the rider is.

5		The “View Cart” bar does not appear as general which is expected to be there when any item is added.	H4 S4		The “View Cart” bar appear when user add any item to the cart to allow them to modified the order.
6		The dropdown button does not appear as obvious and user might not recognise this functionality	H8 S2		Size of button is increased and the “Down” arrow button is change to a filter icon

FOODTM

← Checkout
McDonald UTM Skudai

1 Menu 2 Cart 3 Checkout

Delivery Address

M16, KTD1, Jalan Resak, Johor, 81310

+ Add Delivery instruction

Contactless delivery ☐

Payment Method

Cash RM17.90

Place order

Total (incl. tax) RM17.90

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FOODTM

← Checkout
McDonald UTM Skudai

1 Menu 2 Cart 3 Checkout

Delivery address

M16, KTD1, Jalan Resak, Johor, 81310

+ Add delivery instruction

Contactless delivery ☐

Payment Method

Cash RM13.43

Place order

Total (incl. tax) RM13.43

© HCI Group 4

FOODTM

← Cart
McDonald UTM Skudai

1 Menu 2 Cart 3 Checkout

Delivery Fee: RM3.99

Tax: RM0.03

Before Price: ~~RM17.90~~

First Time User with 25% discount for food and beverage

After Price: RM13.43!

Review Payment and Address

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The above numbering from the restaurant's menu to the checkout does not show the function progress. Users will be confused as to which page they are on.

H3
S3

FOODTM

← Checkout
McDonald UTM Skudai

1 Menu 2 Cart 3 Checkout

Delivery Address

M16, KTD1, Jalan Resak, Johor, 81310

+ Add Delivery instruction

Contactless delivery ☐

Payment Method

Cash RM17.90

Place order

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© HCI Group 4

FOODTM

← Checkout
McDonald UTM Skudai

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FOODTM

← Cart
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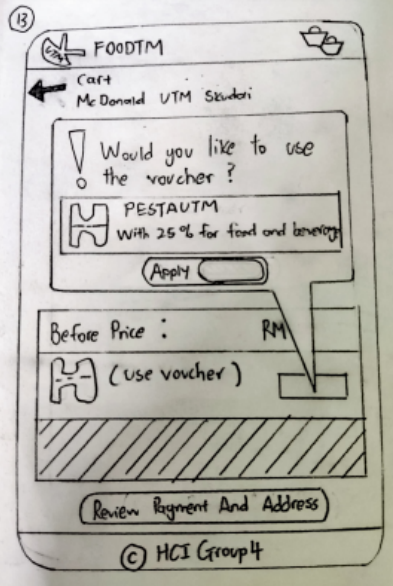
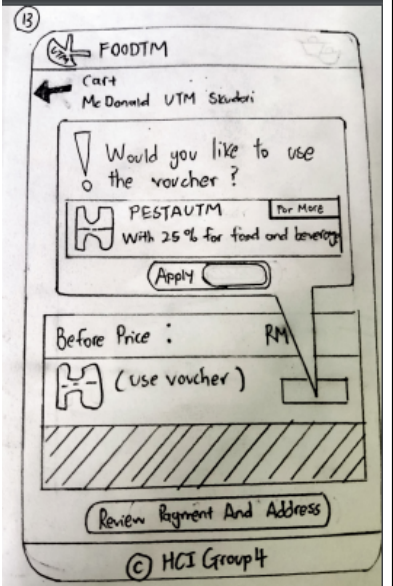
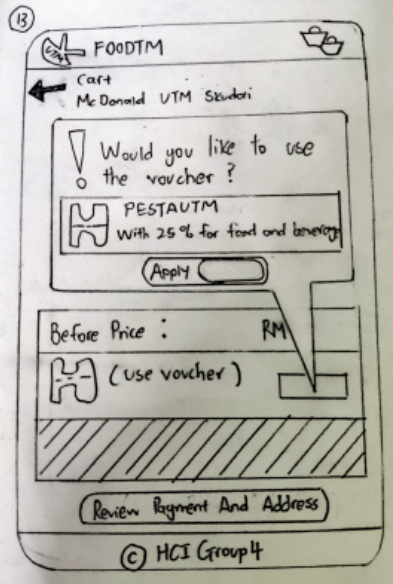
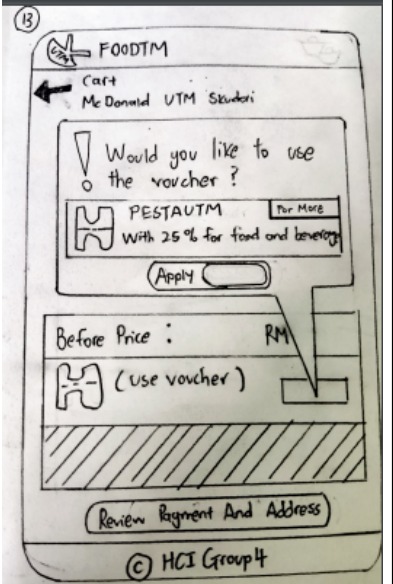
First Time User with 25% discount for food and beverage

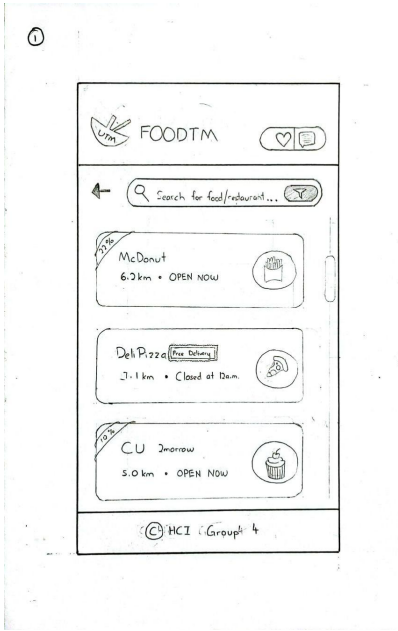
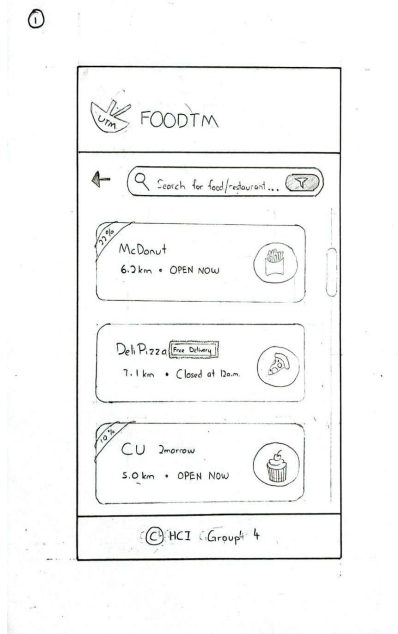
After Price: RM13.43!

Review Payment and Address

© HCI Group 4

The shading on the above line numbering from the restaurant's menu to the checkout will let the user know where they are in the ordering process.

8		Voucher button doesn't seem pressable	H5 S3		A button with the word "For More" has been added.
9		Unnecessary cart icon in cart interface	H3 S2		Remove the unnecessary icon to prevent error

10		Unnecessary buttons appear at the right top of the main menu which might mislead the user to press on it.	H5 C3		Remove the unnecessary button to prevent error
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Reference

1. TOPIC 08 – Other evaluation types
2. Example of running a heuristic evaluation on paper prototype:
<https://www.youtube.com/watch?v=MkyI0gmwfUY>
3. Nielsen's 10 heuristics
4. Visibility of System Status (Usability Heuristic #1)
<https://www.nngroup.com/articles/visibility-system-status/>
5. Maintain Consistency and Adhere to Standards (Usability Heuristic #4)
<https://www.nngroup.com/articles/consistency-and-standards/>